External Documentation

PHASE 1 — Project Scope & Objectives

Define research questions

The project analyzes revenue and profitability trends in the AdventureWorks sales dataset to answer three core business questions:

1. Is the company’s revenue or profitability seasonal?
2. Is there an upward or downward trend across months and years?
3. Which factors (discounts, product categories, quantity, etc.) most affect profitability, and how can it be improved?

Deliverables include:
- A unified analytical data panel (SQL View)
- Data quality and integrity assessment
- Seasonality analysis (monthly & quarterly)
- Trend analysis
- Profitability driver analysis
- Business conclusions and recommendations
- A management-ready presentation

Define metrics (KPIs)

Core performance KPIs
- Total Revenue
- Total Margin (Profit)
- Margin %
- Total Quantity Sold

Time-based KPIs (Seasonality & Trend)
- Monthly Revenue, Margin, Margin %
- Quarterly Revenue, Margin, Margin %
- Yearly Revenue, Margin, Margin %
- Month/Quarter ranking by Margin
- Year-over-Year Revenue and Margin change

Driver-based KPIs (Profitability Analysis)
- Discounts:
  - Average Discount
  - Revenue, Margin, Margin %, Quantity by Discount Bucket
- Product Categories:
  - Revenue, Margin, Margin %, Quantity by Category
  - Category share of total revenue and profit
- Quantity vs Margin:
  - Quantity vs Margin% per Product
  - Volume segment profitability (Low/Medium/High volume groups)

These must be measured monthly and yearly.

Data Sources & Tables Used

The analysis is based on AdventureWorks 2022, using the following tables:
- Sales.SalesOrderHeader — order-level attributes
- Sales.SalesOrderDetail — transaction-level sales data
- Production.Product — pricing & cost
- Production.ProductSubcategory
- Production.ProductCategory

These tables were joined and transformed into a single analytic dataset called General_Panel, containing revenue, profit, margin, discounts, category, and time dimensions.

PHASE 2 — Data Panel Creation & Quality Checks

Data Preparation

A unified analytical view (General_Panel) was created using SQL.
This panel includes:
- Calendar fields (year, month, quarter)
- Product category hierarchy
- Order quantity, pricing, discounts
- Calculated measures:
  - LineRevenue
  - LineCost
  - LineProfit
  - MarginPercent
  - NetUnitPrice
  - UnitMargin

This structure allowed consistent aggregation for all subsequent analyses.

Data Quality Checks

We performed a full integrity review:
- No NULLs in key fields (dates, prices, quantities).
- StandardCost is valid in the panel (0-cost items exist in the Product table but are not sold).
- ~29,000 transactions have negative profit, even without discounts → indicates structural pricing issues (low margins on bikes).
- No duplicate order lines.
- Discounts range 0–40%.
- Product category mapping is complete.

Conclusion: Data is clean enough for analysis, but profit calculations reveal underlying pricing or discounting issues.

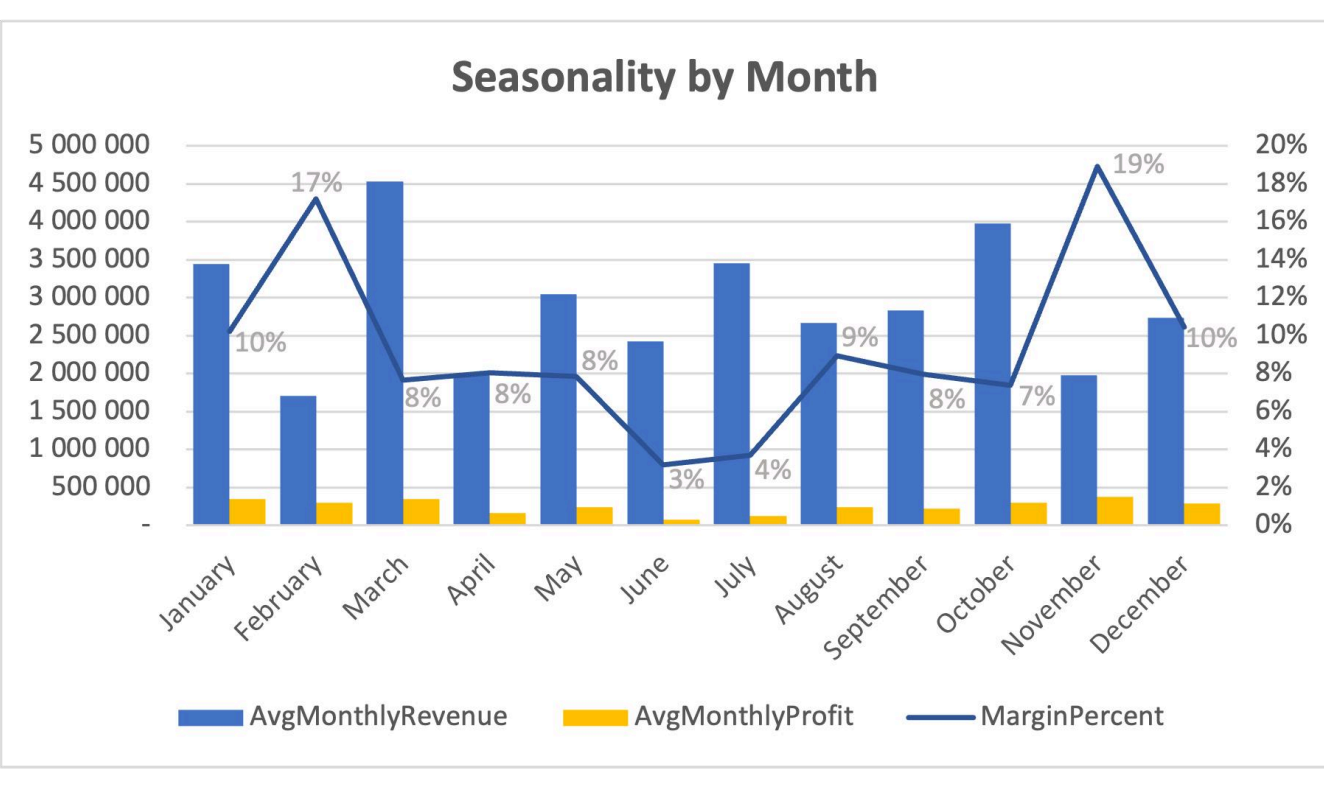
PHASE 3 — Seasonality Analysis

Why We Normalized

Years 2011 and 2014 contain only 8 months and 6 months of data, respectively.
To avoid misleading seasonality conclusions, we:
- Calculated normalized monthly averages (avg revenue & avg profit per year that contains that month).
- Calculated normalized quarterly averages.

This provides a fair comparison across the dataset.

Monthly Seasonality Findings (Normalized)



Revenue Patterns
- Revenue is relatively stable across months.
- Slight peaks in March, October, December.

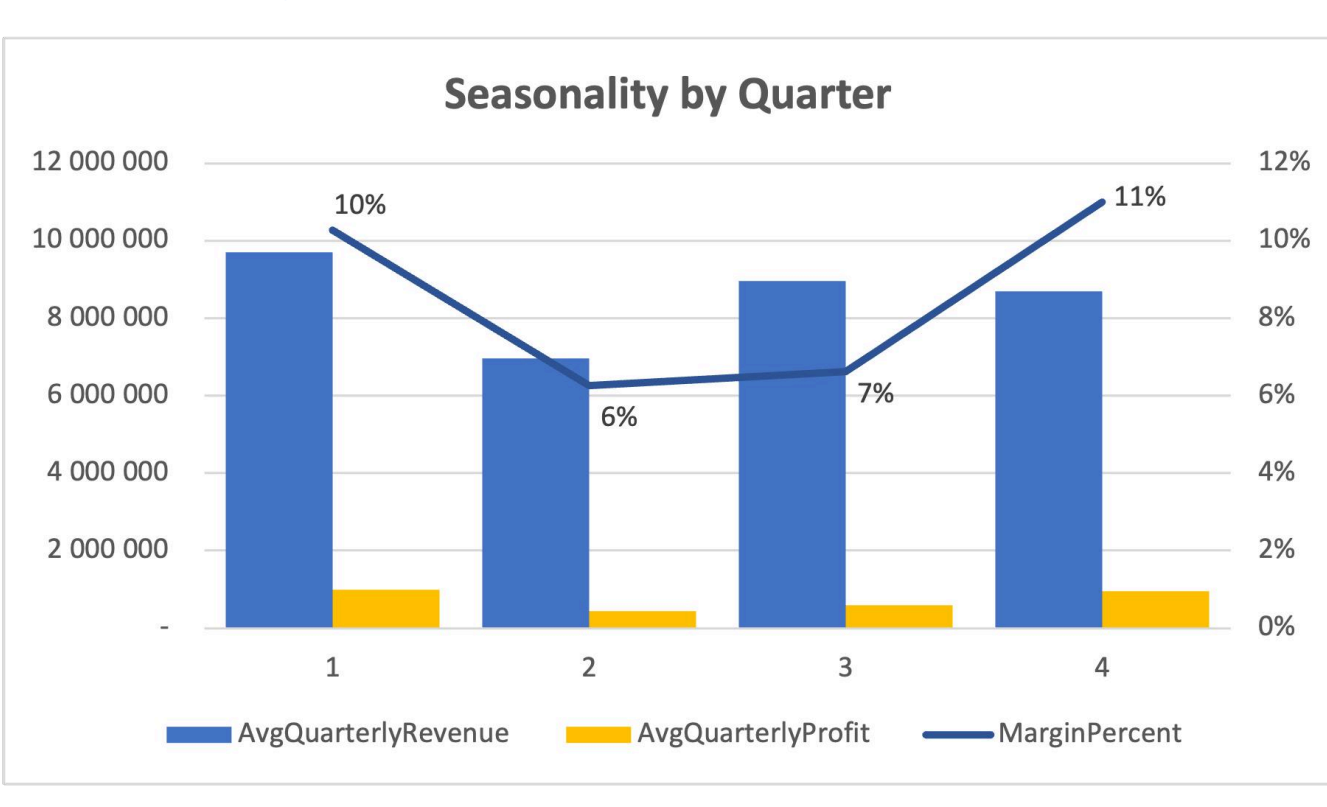
Conclusion: Revenue does not exhibit strong seasonality.
Profitability Patterns

Profitability, unlike revenue, shows clear seasonal behavior:
- Most profitable months:
- November, December, October, August, January, March
- Weakest profitability periods:
- April–June (especially June)

Margin Patterns
Margin % varies dramatically:
- Highest margin:
- November (≈19%)
- Lowest margin:
- June (≈3%), April and July also weak.

Conclusion:
- Profitability is seasonal, with strong Q4 and weak Q2.
- Margin volatility is significant and not explained by revenue → driven by product mix & discounts.

Quarterly Seasonality Findings (Normalized)



Quarterly Conclusions
- Q1 and Q4 are the most profitable periods.
- Q2 consistently underperforms across all years.
- Weak Q2 is responsible for dragging down annual profitability.

Summary of Seasonality Phase
- Revenue is stable year-round with mild peaks.
- Profitability is seasonal, with clear strong (Q1, Q4) and weak (Q2) periods.
- Margin % fluctuations are substantial and largely driven by product mix and pricing issues, not demand.
- Normalization corrected distorted patterns caused by partial years.

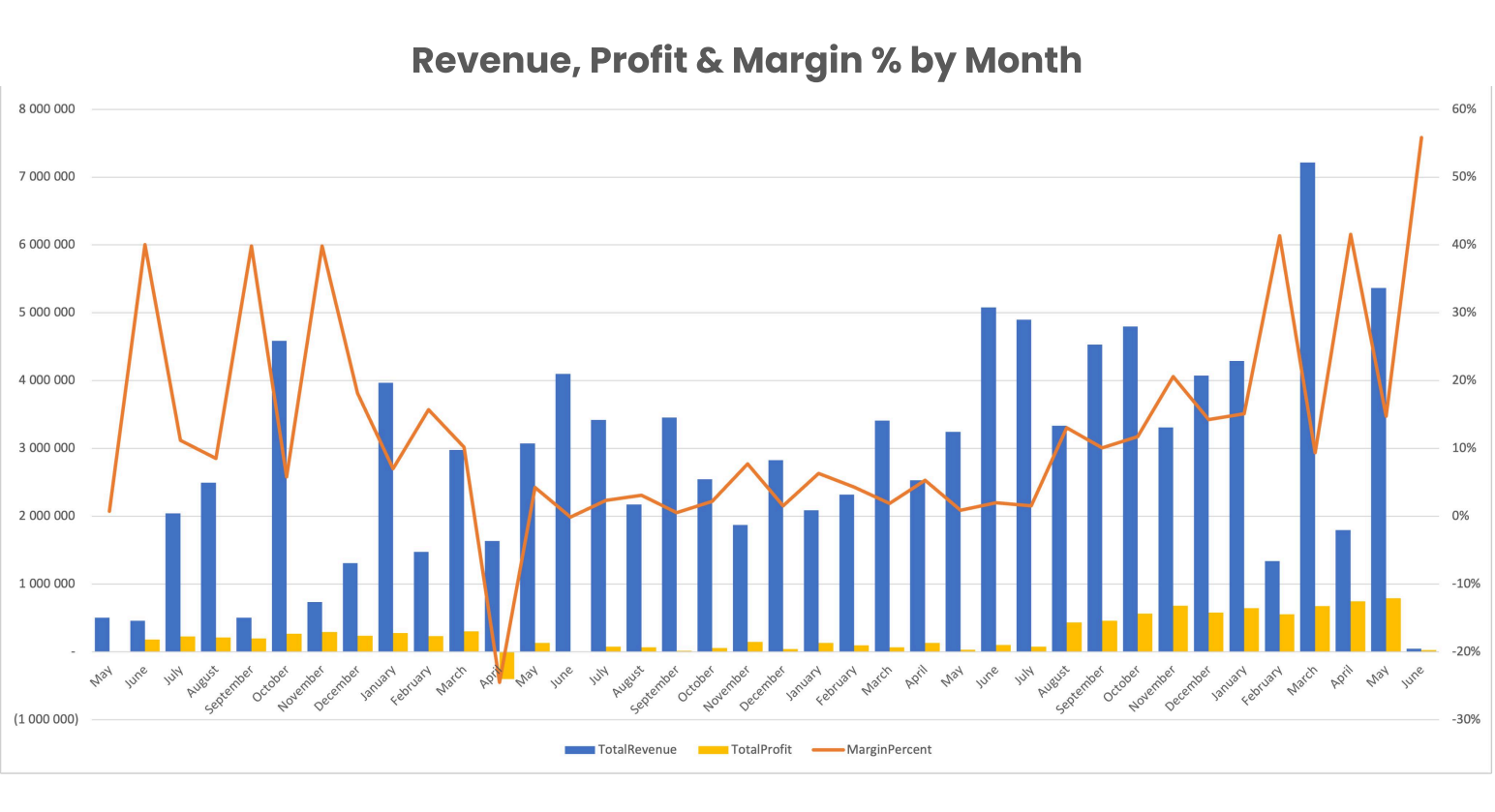
This sets the foundation for the next analysis stages:
- Trend Analysis
- Profitability Driver Analysis (Discounts, Categories, Quantity)

Business Insights From Seasonality

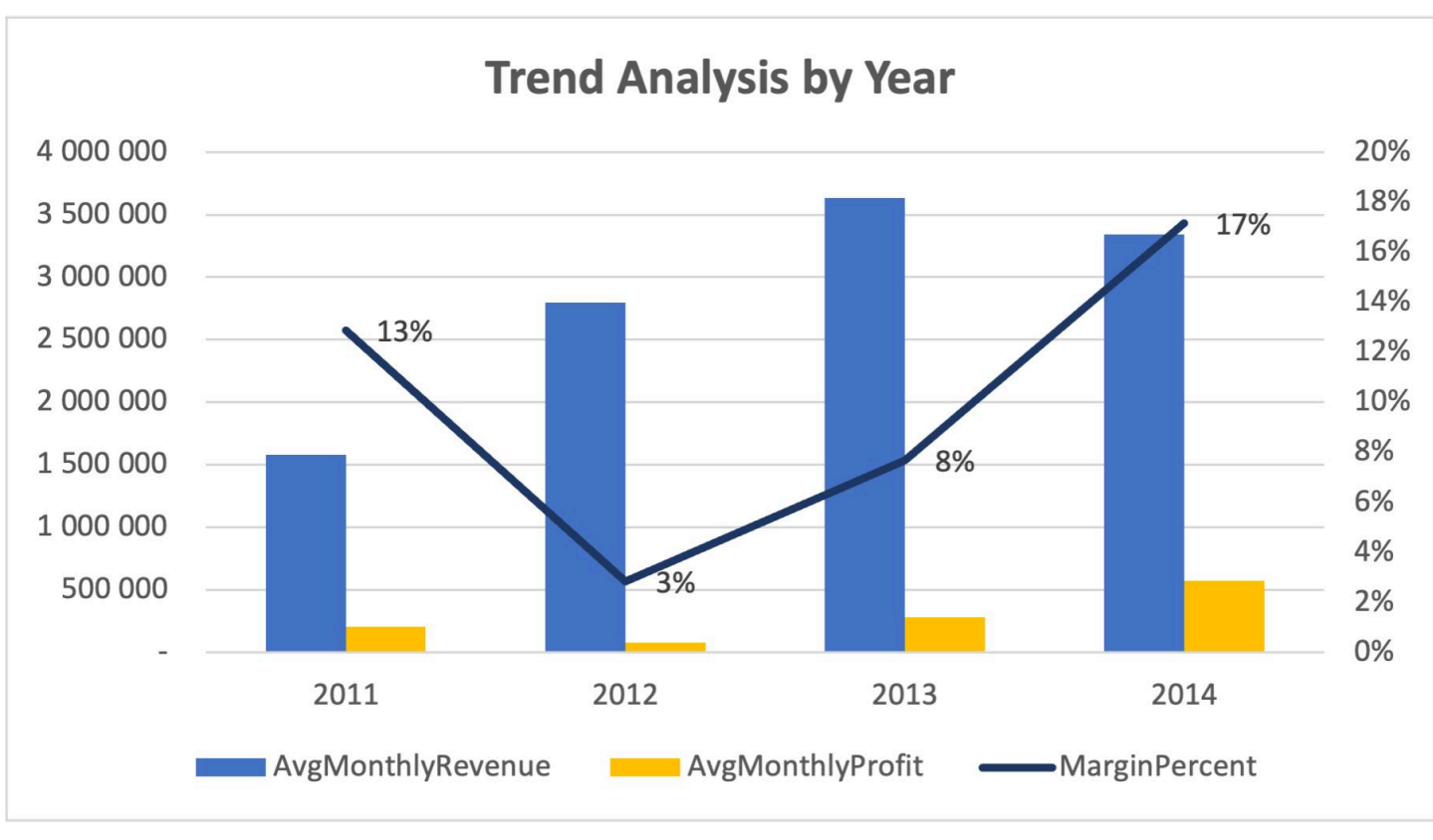
1. Profitability has strong seasonal behavior, unlike revenue.
2. Q2 (April–June) is a structural risk period, driven by low margins, bike-heavy sales, and higher discount sensitivity.
3. Q4 and Q1 represent high-margin windows where the company can push premium products and accessories.
4. Margin % movements correlate strongly with product mix, not overall demand volume.

PHASE 4 — Yearly Trend Analysis

Multi-Year Revenue, Profit, and Margin Trend



Conclusions:
- Revenue grows year-over-year, especially in 2012–2013, showing strong demand expansion.
- Profit does not grow proportionally to revenue, indicating issues in pricing, cost structure, or discounting.
- Margins fluctuate heavily, showing volatility unrelated to seasonality alone.
- Clear seasonal cycles repeat every year: margins rise in Feb, Nov, Dec and fall in Apr–Jun.
- April 2012 is a major anomaly, with extremely negative profitability driven by discount/cost issues.
- 2014 shows margin recovery, suggesting better pricing discipline or corrected product costs.
- High revenue months still produce low margins, proving sales volume is not the main driver of profitability.
- Profitability is highly sensitive to product mix and discounts, not to quantity sold.



Overall, the company is growing in terms profit.

PHASE 5 — Discount & Profitability Analysis

Purpose of This Analysis

The goal of this phase was to understand:
- How discounts affect company profitability
- Whether discounting explains the low or negative margins
- Whether products were sold below cost
- How margin behavior changes across months and years
- Whether quantity or discount is the main driver of low profitability

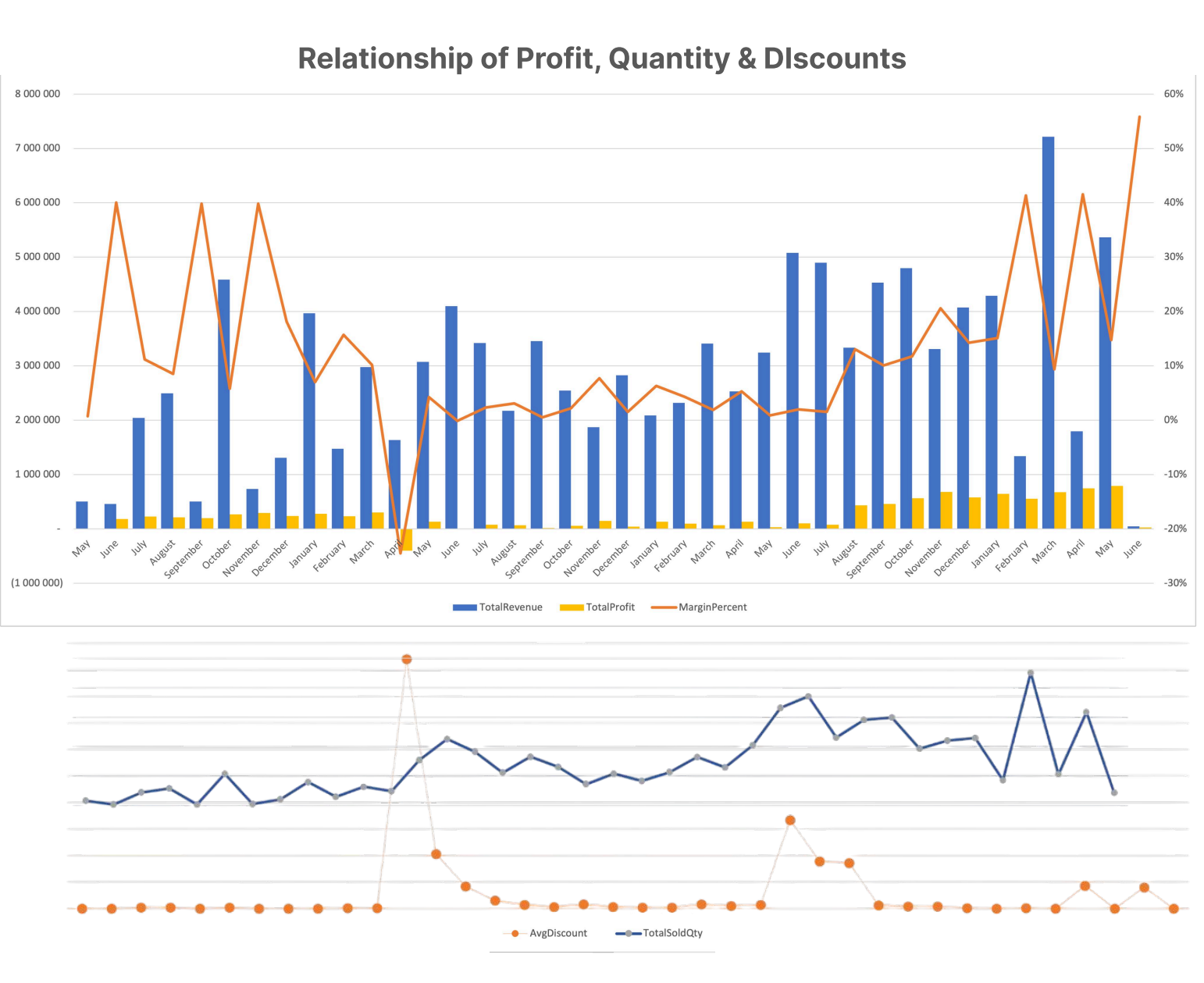
This analysis helps identify structural pricing problems and guides discount policy decisions.

What We Analyzed

1. Discount buckets (0%, 0–10%, 10–30%, 30%+)
2. Discount vs margin trend by month
3. Transactions with negative profit but no discount
4. Per-month metrics including:
   - TotalRevenue
   - TotalProfit
   - MarginPercent
   - AvgDiscount
   - AvgQty
   - Number of negative-profit lines

This allowed us to determine whether discounts, quantity, or pricing errors caused margin volatility.

DiscountBucket	AvgMargin	Revenue	Profit	Transactions
0%	30%	104 423 692	11 264 979	118035
0–10%	7%	4 060 462	- 260 949	2094
10–30%	-74%	1 085 400	- 922 772	882
30%+	-306%	276 827	- 709 354	306



Key Findings

1. Discounts severely damage profitability

Discount bucket results show:
- 0% discount → strong profit (+11M)
- Any discount >0% → margin collapses
- 10–30% discount → heavy losses
- 30%+ discount → extreme losses

➡️ Even small discounts significantly reduce margins.
➡️ Higher discounts are financially catastrophic.

2. Major pricing issue: many sales have no discount but negative profit

Across almost every month:
- Thousands of transactions with 0% discount
- But LineProfit < 0
- Meaning UnitPrice < StandardCost

This reveals a structural pricing or data configuration problem, not a discounting strategy issue.

This problem is especially severe in:
- 2012 (multiple months with thousands of below-cost sales)
- Early 2013
- Some periods in 2014

➡️ The system allowed selling below cost without recording any discount.
➡️ This is the biggest cause of profit leakage.

3. Quantity does NOT explain profitability

Analysis showed:
- AvgQty is stable (1–3 units)
- No consistent relationship between quantity and margin
- Discounts and incorrect pricing are the real drivers of loss

D. 2012 is the worst year

2012 shows:
- Very low margins
- Multiple months with negative total profit
- Thousands of negative-profit lines every month
- Some discounts applied, but most losses come from incorrect pricing

➡️ Pricing logic or cost updates were likely incorrect in 2012.

4. Margin improved significantly in late 2013–2014

As pricing stabilizes:
- Margins naturally rise
- Even small discounts do not cause extreme losses
- Negative-profit transactions decrease

This confirms that pricing logic corrections, not lower discounts, improved performance.

Summary

The discount analysis revealed that while discounts significantly reduce profitability, the main driver of margin loss is a pricing system that allowed selling products below cost even without a recorded discount. This issue affected tens of thousands of transactions, especially in 2012–2013. Discounts above 5% amplify the losses, but they are not the root cause. To improve profitability, the company needs strict pricing controls, discount limits, and automated monitoring to prevent below-cost sales and track margin performance more effectively.

Business Conclusions

1. The main problem is NOT discounting — it is pricing below cost.

Discounts make losses worse, but the root cause is:
- UnitPrice < StandardCost
- No discount recorded
- Thousands of lines sold at guaranteed loss

This cannot happen in a healthy pricing system.

2. Discounts above ~5% should be heavily restricted

Based on profit behavior:
- 0–5% → manageable
- 5–10% → risky
- 10–30% → destroys profit
- 30%+ → unacceptable

3. Cost and price updates must be reviewed

2012 shows evidence of:
- Wrong standard costs
- Incorrect pricing rules
- System configuration issues

4. Introduce guardrails into the pricing system
- Do NOT allow UnitPrice < StandardCost
- Block or warn when LineProfit < 0
- Require approval for discounts above 5%
- Require approval for bulk discounts
- Track the number of negative-profit transactions per month

5. Build monthly monitoring dashboards

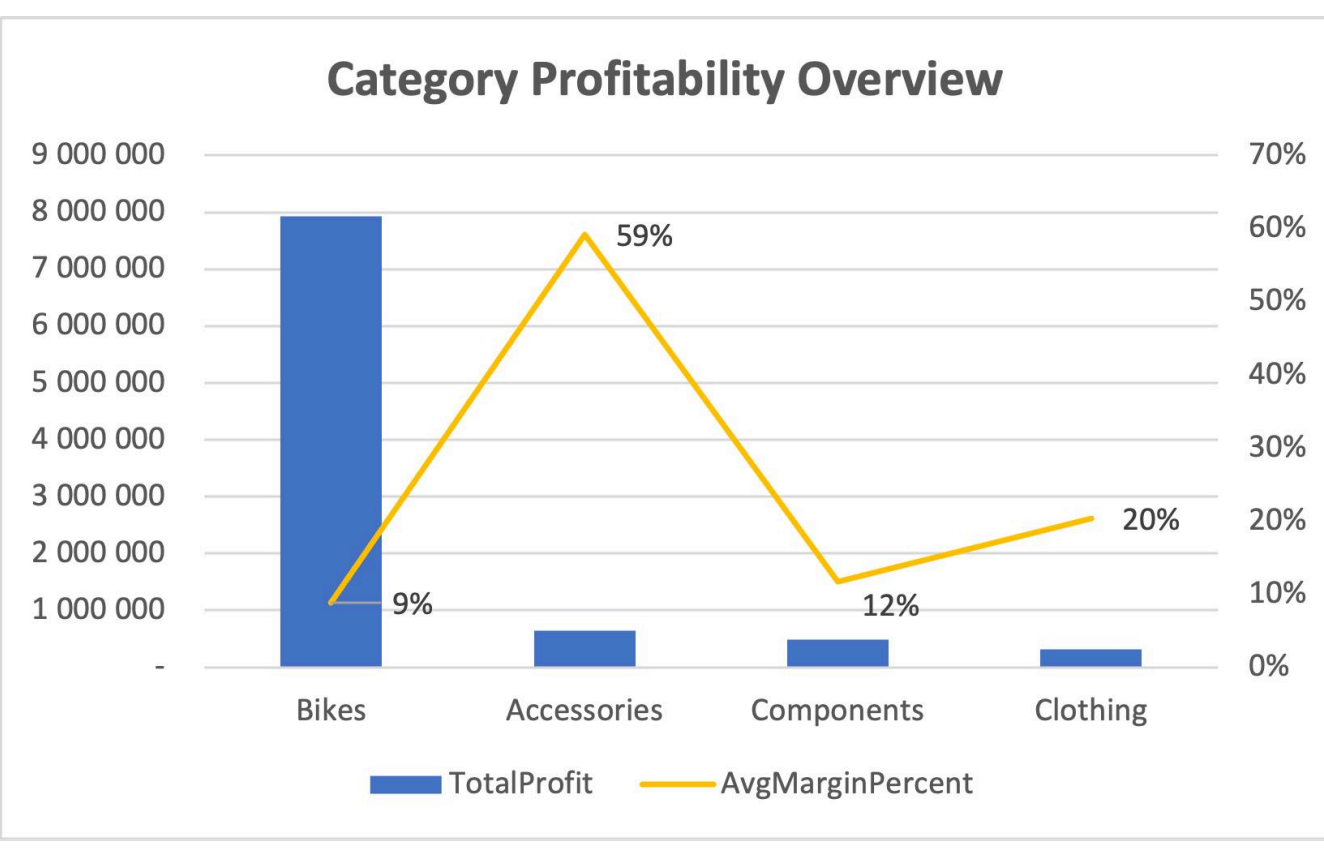
To avoid future leakage, track:
- Margin %
- Discount usage
- Negative-profit transactions
- Price–cost mismatch alerts
- Category profitability

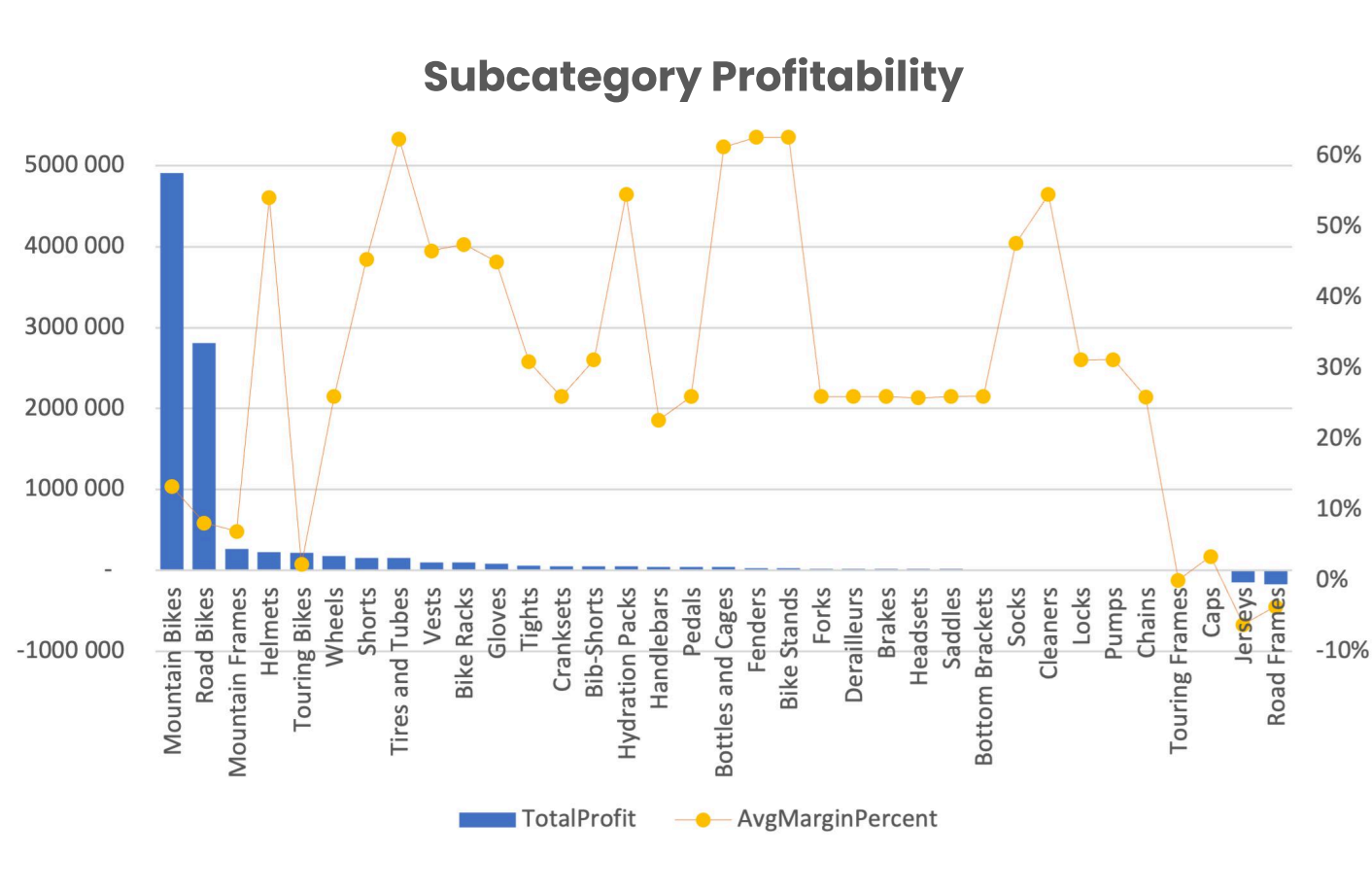
PHASE 6 — Category & Product Profitability Analysis

What We Analyzed

In this phase, we examined profitability across:
- Product Categories (Bikes, Components, Clothing, Accessories)
- Product Subcategories (e.g., Mountain Bikes, Road Frames, Helmets)
- Individual Products (top profit generators and loss makers)

What We Discovered





Top 20 performing products are: (according to profit)

1 Mountain-200 Black, 42
2 Mountain-200 Black, 38
3 Mountain-200 Black, 46
4 Mountain-200 Silver, 38
5 Mountain-200 Silver, 46
6 Mountain-200 Silver, 42
7 Road-150 Red, 48
8 Road-150 Red, 62
9 Road-150 Red, 52
10 Road-150 Red, 56
11 Road-150 Red, 44
12 Touring-1000 Blue, 54
13 Road-250 Red, 58
14 Road-250 Red, 48
15 Road-350-W Yellow, 44
16 Touring-1000 Blue, 50
17 Touring-1000 Blue, 46
18 Road-350-W Yellow, 42
19 Road-250 Red, 52
20 Hitch Rack - 4-Bike

Loss maker products are 60: (according to profit)

1 Road-650 Red, 44
2 Touring-1000 Yellow, 60
3 Road-650 Red, 60
4 Touring-1000 Yellow, 46
5 Road-650 Black, 52
6 Long-Sleeve Logo Jersey, L
7 Road-650 Red, 62
8 Road-650 Black, 58
9 Road-650 Red, 48
10 Touring-3000 Blue, 50
11 Touring-3000 Yellow, 62
12 Road-250 Black, 44
13 Touring-3000 Yellow, 44
14 ML Road Frame-W - Yellow, 44
15 Short-Sleeve Classic Jersey, XL
16 HL Road Frame - Red, 62
17 HL Road Frame - Red, 44
18 Touring-3000 Blue, 54
19 Touring-3000 Yellow, 50
20 Long-Sleeve Logo Jersey, M
21 LL Road Frame - Black, 52
22 HL Road Frame - Black, 44
23 Road-550-W Yellow, 48
24 Short-Sleeve Classic Jersey, L
25 ML Road Frame-W - Yellow, 38
26 ML Road Frame-W - Yellow, 48
27 Long-Sleeve Logo Jersey, XL
28 Road-650 Red, 52
29 Road-650 Black, 44
30 Road-550-W Yellow, 38
31 LL Road Frame - Black, 58
32 Road-650 Black, 60
33 Touring-3000 Yellow, 54
34 Touring-3000 Blue, 58
35 Short-Sleeve Classic Jersey, S
36 Road-450 Red, 52
37 Road-450 Red, 58
38 AWC Logo Cap
39 ML Road Frame-W - Yellow, 40
40 HL Road Frame - Black, 48
41 HL Road Frame - Red, 48
42 Touring-3000 Yellow, 58
43 LL Road Frame - Black, 44
44 Road-450 Red, 60
45 Road-450 Red, 44
46 Road-650 Black, 48
47 HL Touring Frame - Blue, 54
48 HL Touring Frame - Yellow, 54
49 LL Road Frame - Black, 60
50 Road-450 Red, 48
51 HL Touring Frame - Blue, 60
52 Touring-3000 Blue, 44
53 Road-650 Black, 62
54 HL Touring Frame - Yellow, 60
55 LL Touring Frame - Yellow, 62
56 LL Touring Frame - Yellow, 44
57 LL Touring Frame - Blue, 58
58 LL Touring Frame - Blue, 62
59 LL Touring Frame - Blue, 44
60 LL Touring Frame - Yellow, 58

Key Conclusions

1. Bikes dominate total company profit
- ~80% of total profit comes from Bikes, especially:
  - Mountain Bikes (highest profit & margin consistency)
  - Road Bikes

These categories are the financial backbone of the business.

2. Accessories deliver excellent margins

Even though volume is moderate, Accessories show:
- Very high margins (50–60%)
- Low operational issues
- Stable profitability across all periods

Top performers include Helmets, Bike Racks, Tires & Tubes.

This category is small but extremely efficient — an opportunity for growth.

3. Components are profitable overall but contain structural loss-makers
- Many component types are profitable (Wheels, Cranksets, Derailleurs),
- But Road Frames and Touring Frames consistently generate losses.
- Frames appear as both theoretical and actual loss-makers, meaning:
  - The system price is below cost → structural pricing issue, not discounts.

This is a major root cause of negative lines.

4. Clothing is the weakest-performing category
- Thin margins overall
- Several subcategories are persistent loss-makers, especially:
  - Jerseys
  - Caps

Clothing behaves as a low-margin, high-discount, high-negative-line segment.

5. Loss-making products are highly concentrated

Most loss comes from:
- Road-650 variants
- Touring-1000 / Touring-3000 variants
- Several Road Frame SKUs
- Key Clothing SKUs (Logo Jerseys)

The pattern is consistent across years → this is not random.

Recommendations

1. Immediate pricing review for loss-making SKUs

Especially:
- Road & Touring Frames
- Road-650 bike variants
- Clothing Jerseys
- These products should be:
- Repriced,
- Bundled differently,
- Or removed if they cannot achieve margin neutrality.

2. Expand and promote high-margin categories
- Accessories (Helmets, Packs, Tires, Racks)
- Mountain Bikes (industry-best margin + volume combination)

These products improve profitability without heavy discounting.

3. Reduce discounts in categories with already low margins

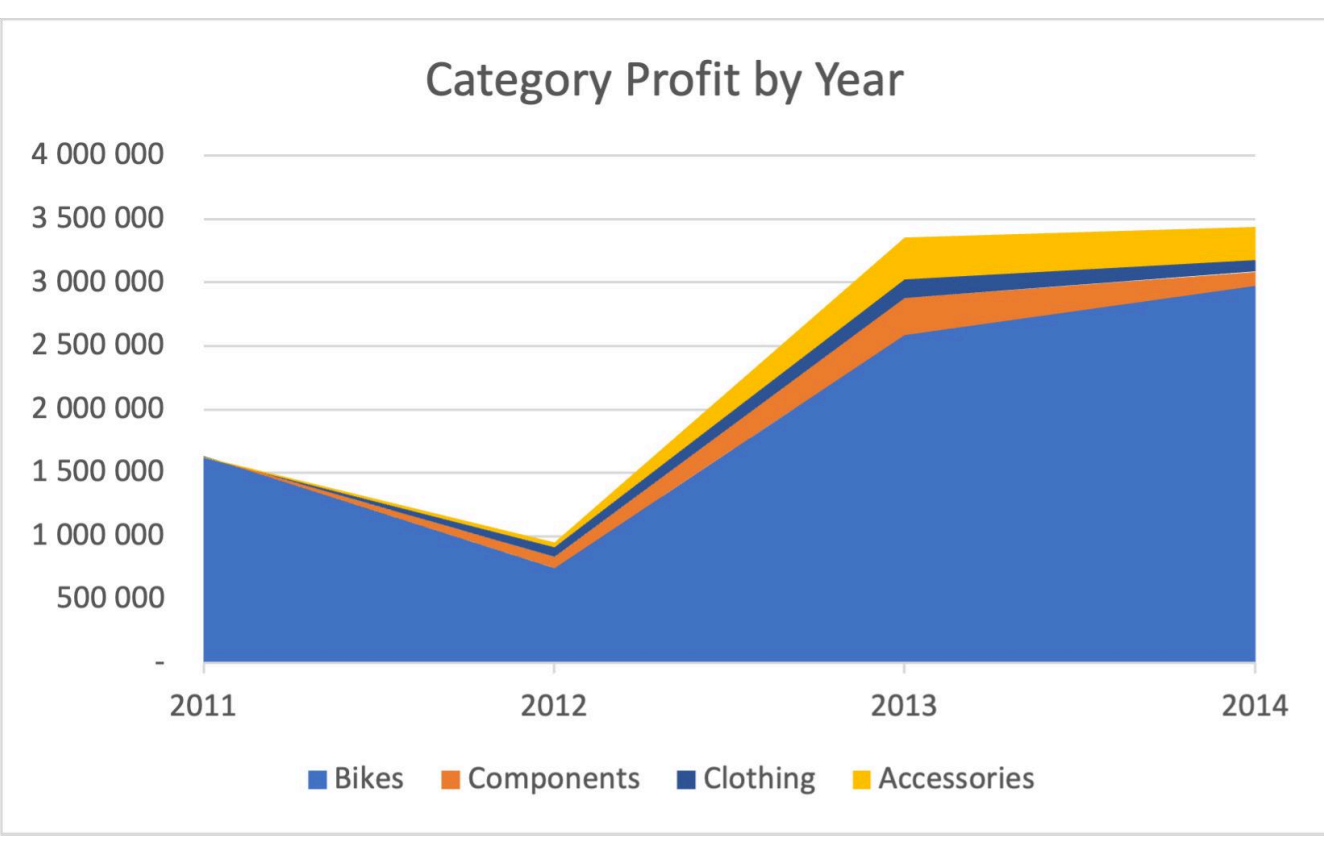
Clothing and certain Components cannot tolerate discounts —
each discount pushes them deeper into loss.

4. Category Mix Monitoring

Monthly/quarterly category mix should be tracked because:
- Shifts toward Clothing / Road Frames significantly lower total margin,
- Shifts toward Bikes & Accessories lift profitability.

5. Use the Category Mix chart for operational decision-making

Management can clearly see:
- Which categories drive profit,
- Which categories reduce margin,
- How category composition explains sudden margin swings.



The Area Chart illustrates how each major product category contributed to total annual profit over time, using color-coded layers to show their relative weight. This visualization makes it easy to see shifts in category mix: Bikes consistently dominate profit, Accessories steadily grow as a high-margin segment, while Components and Clothing show smaller and more volatile contributions. The stacked areas highlight not only the absolute profit of each category but also how changes in product mix directly influence overall profitability year to year.

Products With Negative Profit Lines – Root Cause

What we did
- Filtered all order lines with negative LineProfit and, for each SKU, compared:
- StandardCost, ListPrice, theoretical margin, actual average selling price, and realized margin%.

Key findings
- Products are profitable at list price. For almost all “problem” SKUs, ListPrice – StandardCost is clearly positive with healthy theoretical margins (often 23–45%+).
- Losses come from how we sell, not what we sell. Many SKUs are actually sold on average below StandardCost, so realized margins become negative (e.g. Road-650 / Road-550-W / Road-250 families, Touring-1000/3000, Mountain-100/500 Silver, logo jerseys, caps, and several frames).
- Some frames show small but systematic negative margin, suggesting minor price overrides or misaligned price lists rather than bad cost data.

Conclusions & recommendations
- The portfolio itself is economically sound, but pricing execution is leaking margin on specific SKUs.
- Implement price floors (no sale below StandardCost without explicit approval) and system alerts when NetUnitPrice < StandardCost.
- Create a “loss-maker watchlist” (problem SKUs) and review their realized margins monthly to quickly catch future pricing or configuration errors.

PHASE 7 — Final Summary, Conclusions & Recommendations

Discounts & Margin Performance

What we learned:
- 0% discount sales drive almost all profit (~$11.2M).
- 10–30% and 30%+ discounts destroy margin, producing large negative profit contributions.
- Monthly margin swings closely follow how much discounted volume enters the mix, not changes in cost or demand.

Conclusion:
Discounting is the #1 margin leak in the business.

Recommendation:
- Set strict discount governance (caps, approvals, price floors).
- Monitor discount buckets weekly (0%, 0–10%, 10–30%, 30%+).
- Eliminate unprofitable discount levels.

Category & Subcategory Profitability

What we learned:
- Bikes generate ~80%+ of total profit.
- Accessories are small but very high-margin contributors.
- Clothing contributes margin but contains several consistent loss-makers.
- Subcategories show extreme concentration:
  - Mountain Bikes and Road Bikes generate most profit.
  - Jerseys, Caps, Road Frames, Touring Frames are persistently negative.

Conclusion:
Profit comes from a narrow set of winning subcategories, while several smaller categories dilute profitability.

Recommendation:
- Protect inventory and availability of high-performing product families.
- Reduce SKUs or fix pricing for low-performing subcategories.
- Implement category mix dashboards to understand monthly margin swings.

Category Mix Over Time (Area Chart Insight)

What we learned:
- Yearly profit increases correlate directly with higher share of Bikes in the mix.
- When the mix shifts toward Components or Clothing, profit margins fall sharply (2012 dip).
- Profitability is therefore mix-driven, not seasonality-driven.

Conclusion:
Strategic focus must remain on optimizing category contribution, not chasing volume everywhere.

Recommendation:
- Set target mix ratios (e.g., Bikes ≥ 70% of revenue).
- Monitor mix monthly using a stacked area chart.

Product-Level Profitability

What we learned:
- The top 100 most profitable SKUs are almost entirely Bikes + selected Accessories + selected Clothing.
- Bottom 60 SKUs show consistent losses, primarily in:
  - Road-650 / Road-550 / Road-250 families
  - Touring series
  - Logo Jerseys & Caps
  - Road and Touring Frames

Conclusion:
Massive concentration: a small product subset generates most profit, while dozens of SKUs create recurring losses.

Recommendation:
- Build a “core SKU” list to prioritize, promote, and protect.
- Build a “loss SKU” audit list to fix or discontinue.

Root Cause of Negative Profit Lines

What we learned:
- Theoretical margins are healthy for almost all products.
- Products become unprofitable because actual selling prices fall below StandardCost.
- This is a pricing execution failure, not an economics problem.

Conclusion:
The business loses money not because products are bad, but because discounting and overrides undercut profitability.

Recommendation:
- Introduce a system rule: “No sale below cost” without manager override.
- Implement automated alerts when UnitPrice < StandardCost.
- Review pricing policies for problem families monthly.

Final Conclusions

1. Profitability is overwhelmingly driven by discount discipline and category mix.
2. The product catalog is fundamentally strong, but pricing execution destroys margin.
3. A small number of SKUs drive most profit; a long tail destroys value.
4. Operational controls, not demand changes, will unlock major profit improvement.

Final Recommendations

High Priority
- Enforce discount caps and price floors.
- Create dashboards for discount bucket performance and category mix.
- Fix pricing for the loss-maker SKUs (Road-650, Touring frames, Jerseys, Caps).

Medium Priority
- Rationalize the catalog: reduce or reprice unprofitable variants.
- Promote high-margin Accessories (Helmets, Racks, Tires).
- Implement category profitability targets for managers.

Low Priority
- Optimize seasonal purchasing based on mix trends.
- Rebuild long-term pricing strategy (good/better/best structure).